Enhancing LoRa Reception with Generative Models: Channel-Aware Denoising of LoRaPHY Signals

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Abstract

The proliferation of Internet of Things (IoT) applications relying on Low Power Wide Area Networks (LPWANs) demands robust and energy-efficient communication solutions. Among various LPWAN technologies, LoRa emerges as a prominent choice due to its long-range capabilities and low energy consumption. However, the practical deployment of LoRa is hindered by significant signal degradation caused by channel and hardware noise, especially in urban environments. We introduce *GLoRiPHY*, a novel generative framework designed to enhance the reception quality of LoRaPHY signals through a channel-aware denoising mechanism. Utilizing a transformer-based architecture, *GLoRiPHY* leverages the known preamble of LoRaPHY signals to compensate for channel-induced distortions, thereby generating a clean signal suitable for direct demodulation. The system integrates Convolutional Neural Networks (CNNs) for efficient feature encoding and decoding, maintaining a compact model footprint even at higher Spreading Factors (SFs). Evaluations on real-world and simulated datasets show that in comparison to the current state-of-the-art solution, *GLoRiPHY* significantly lowers the Symbol Error Rate (SER) by up to **2.85x** and demonstrates generalizability in unseen environments, while reducing inference times by up to **5.75x**.

CCS Concepts

• **Networks** → *Sensor networks*; **Error detection and error correction**; • **Computing methodologies** → **Neural networks**.

Keywords

LoRa, Error Correction, Signal Denoising, Deep Learning

ACM Reference Format:
Kanav Sabharwal, Soundarya Ramesh, Jingxian Wang, Dinil Mon Divakaran, and Mun Choon Chan. 2024. Enhancing LoRa Reception with Generative Models: Channel-Aware Denoising of LoRaPHY Signals. In *ACM Conference on Embedded Networked Sensor Systems (SenSys '24), November 4–7, 2024, Hangzhou, China.* ACM, New York, NY, USA, 14 pages. https://doi.org/10.1145/3666025.3699354

SenSys '24, November 4–7, 2024, Hangzhou, China

ACM ISBN 979-8-4007-0697-4/24/11
https://doi.org/10.1145/3666025.3699354

1 Introduction

Recent years have witnessed the widespread adoption of Low Power Wide Area Networks (LPWANs) to provide wireless connectivity to battery-constrained Internet of Things (IoT) devices over distances typically in the range of kilometers. Applications span structural monitoring [47], smart agriculture [43], smart metering systems [6], and monitoring at construction and mine sites [22]. Among available LPWAN technologies such as LoRa [27], SIGFOX [4], and NB-IoT [5], LoRa remains a favored choice due to its open-source nature and operation over unlicensed frequency bands. LoRa technology promises communication ranges of several kilometers and battery lives of 5-10 years at a low power cost.

However, studies [7, 18, 25, 29, 38] reveal significantly increased packet corruption, particularly in urban environments. The high rate packet corruption in such scenarios can be attributed to factors such as strongly attenuated signals, multi-path fading, and cheap hardware noise, which cause frequency and timing offsets. Common countermeasures include using highly redundant Hamming codes (up to one redundant bit for every transmitted bit, i.e., 4/8), increasing transmit power levels, or utilizing higher Spreading Factors (SFs)—a network configuration that nearly doubles the time on air with each increase in SF. However, these measures lead to higher power consumption at the radio front end, significantly reducing the battery life of the end nodes.

Previous works [2, 8, 10, 12, 13] have utilized MIMO-based techniques to counteract channel effects in scenarios with densely deployed LoRa nodes and gateways. These techniques use signals transmitted from multiple nodes or received at multiple gateways to collaboratively enhance signal demodulation/decoding, effectively mitigating the channel effects.

NELoRa [23], on the other hand, introduced an ML-based approach, demonstrating similar gains even with a single node/gateway by leveraging AI capabilities for demodulation. However, NELoRa has notable limitations: 1) Its design does not account for multi-path noise variation across different environments, limiting its generalization capability in *unseen* deployment setups. 2) It employs a masking-based approach to *ignore* Gaussian noise, which leads to spectral loss and renders the processed signal unsuitable for direct LoRa demodulation. This necessitates the use of classification models to identify up to 4096 possible symbols, thereby requiring substantial memory and computing power, and incurring high inference times. Our goal is to address these shortcomings of the current state-of-the-art ML-based solutions.

We introduce *GLoRiPHY* [1] (Generative framework for LoRaPHY Improvement) – a novel approach to enhance the reception quality of LoRa signals through a channel-aware denoising mechanism. Specifically, we develop a generative framework that generates channel-compensated payload symbols by utilizing the preamble of LoRaPHY signals to determine how the channel modifies the received signal. *GLoRiPHY* is designed to effectively process the received preamble, extracting relevant features regarding how the channel impacts the signal in terms of time delay, amplitude attenuation, and phase shifts. These features are then utilized to generate a cleaned payload symbol, compensated for channel effects.

The symbol generation process within *GLoRiPHY*'s generative model is also designed to enable direct processing by the standard LoRa demodulation algorithm, thus eliminating the need for bulky classification models to identify processed symbols.

Through extensive evaluations on real-world and simulated datasets, we demonstrate the capability of *GLoRiPHY* to enhance reception quality in terms of a lower Symbol Error Rate (SER), better generalization capability in unseen scenarios, and significantly reduced inference time.

We summarize our contributions:

- To the best of our knowledge, *GLoRiPHY* is the first machine learning-based, channel-aware signal denoising framework specifically designed for LoRaPHY. This approach represents a significant advancement in applying machine learning techniques to improve long-range communication technologies, allowing generalisability in unseen environments.
- We implement strategies for efficient feature extraction and management to maintain models with low computational requirements. This ensures that *GLoRiPHY* remains practical and effective across various network configurations, making it suitable for deployment in a wide range of environments.
- We test *GLoRiPHY* against the current state-of-the-art (SoTA) on both real-world and simulated datasets. Our evaluations demonstrate Symbol Error Rate (SER) gains ranging from 0.68% to 16.3% and inference time reductions ranging from 2.64x to 5.75x. These results not only showcase the efficacy of *GLoRiPHY* in enhancing signal quality but also its efficiency in processing, which are critical for real-time applications.

2 Related Work

We present prior work on improving LoRa reception.

Channel Equalization. Traditional channel equalization techniques, often effective in other communication systems, face specific challenges in LoRaPHY due to its unique modulation scheme and the non-linearities introduced by the use of cheap hardware in the form of frequency and phase offsets.

A recent study attempts to adapt these techniques using LoRa's preamble to compensate for channel effects [34]. However, their evaluation makes several simplistic assumptions such as knowledge of the channel covariance matrix and a constant delay profile for the channel, both of which do not hold in practice.

MIMO-Based Approaches. Several studies have explored compensation of channel effects in LoRa by leveraging dense deployments of nodes and gateways for collaborative demodulation and decoding. The key idea behind such approaches is to utilise multiple copies of the same signal received at multiple gateways/transmitted by multiple nodes, to compensate for the channel noise.

Charm [8] demonstrates improvements by coherently combining signals received at multiple gateways. Chime [12] optimizes this approach by adapting the frequency configuration best suited for a given environment by modeling strong reflectors in multipath reception. Choir [10] and QUAIL [13] leverage collaborative transmissions from multiple nodes to enhance signal transmission. OPR [2] recovers corrupted packets by capitalizing on disjointly decoded bit errors from multiple gateways, each experiencing unique channel conditions.

These MIMO-based solutions offer improvements in LoRa reception by addressing complex channel noise, including multipath distortions. However, their efficacy relies heavily on the availability of densely deployed nodes or gateways, which is not feasible in many deployment scenarios.

ML-Based Approaches. Machine learning has shown promise in enhancing demodulation capabilities within wireless networks [19, 28, 30, 35, 45, 49]. In the context of LoRa, DeepLoRa [26] uses machine learning to estimate path loss based on land cover data, facilitating informed network deployment. ChirpTransformer [24] introduces a novel ML-based encoding scheme that uses fine-grained chirp features to tailor encoding scheme to specific deployments.

NELoRa [23] utilizes a DNN-based approach to improve LoRa demodulation, achieving performance comparable to the MIMO-based methods without requiring collaboration among nodes. NELoRa introduces some of the key enablers in applying ML to enhance LoRa demodulation. However, NELoRa's design, primarily assumes AWGN conditions, limiting its generalizability across varied real-world scenarios. The approach also relies on spectral masking to *ignore* noise, which can result in significant energy loss, making it incompatible with traditional LoRa demodulation techniques and necessitating a DNN-based classifier for signal identification.

We address the shortcomings of existing ML-based solutions by integrating channel context into *GLoRiPHY*, effectively compensating for channel effects. Our approach not only enhances the capabilities to denoise the received signal but also ensures low inference times and full compatibility with standard LoRa demodulation techniques. Thus, *GLoRiPHY* promises broad applicability and improved performance across diverse operational environments.

3 Background and Motivation

3.1 LoRaWAN and LoRaPHY

LoRa (Long Range) is a modulation technology designed for low-power, long-range communication, operating in sub-GHz unlicensed frequency bands. It employs Chirp Spread Spectrum (CSS) modulation, which modulates data as a series of chirp signals that continuously vary in frequency [25].

The Spreading Factor (SF) is a key network configuration in LoRa that determines the number of data bits encoded by a single chirp signal. For instance, at SF8, there are $2^{\mathrm{SF}} = 256$ possible symbols, each encoding 8 bits. These symbols are uniformly spaced in frequency, initiating each chirp at a unique frequency increment. SF values range from 7 to 12, with each increment doubling the time

[1] The source code, along with the dataset and documentation, is available at https://github.com/KanavSabharwal/GLoRiPHY

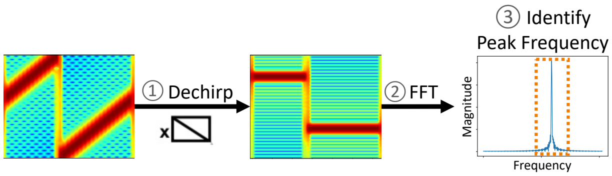


Figure 1: LoRa's demodulation process – performing the *dechirp* operation, transforming the signal to the frequency domain via Fast Fourier Transform (FFT), and identifying the frequency with the largest magnitude.

on air for each symbol, thus enhancing the communication range but at the cost of reduced bit rate and increased power consumption.

LoRa Demodulation Although the LoRa physical layer, or LoRaPHY, remains proprietary to Semtech, various studies [20, 40, 48] have reverse-engineered it, enabling further research. We outline the LoRa demodulation process in Figure 1. The steps involved in demodulating an incoming signal $e^{-j2\pi f_{\text{carrier}}t}$ are detailed below:

1. *CFO and TO Correction*: The signal undergoes Carrier Frequency Offset (CFO) and Timing Offset (TO) during transmission due to inaccuracies in hardware frequency clocks. These offsets are mitigated based on estimations derived from the preamble:

$$s_{\text{corr}}(t) = e^{-j2\pi(f_{\text{carrier}}+\Delta f_{\text{CFO}})t} + \tau_{\text{TO}}$$

Here, Δf_{CFO} and τ_{TO} represent the estimated frequency and timing offsets, respectively.

2. *Dechirp*: The corrected signal is multiplied by a base downchirp to align the frequency components:

$$s_{\text{dechirp}}(t) = s_{\text{corr}}(t) \cdot e^{-j2\pi\left(f_{\text{carrier}}+\frac{B}{2}-\frac{B}{T}t\right)t}$$

Here, B is the chirp's bandwidth, and T its duration.

3. *FFT*: The Fast Fourier Transform (FFT) is applied to the dechirped signal to convert it into the frequency domain. The FFT is typically oversampled to enhance frequency resolution:

$$S_{\text{FFT}}(k) = \text{FFT}(s_{\text{dechirp}}(t))$$

4. *Peak Searching*: The FFT output is analyzed to identify the frequency with the highest amplitude, which corresponds to the original chirp frequency.

Although we present an ideal scenario with a clear peak visible in Figure 1, real-world conditions often involve significant signal attenuation and channel fading. These conditions complicate the accurate identification of the peak, leading to packet corruptions from incorrectly identified symbols. In the following section, we explore how channel noise impacts RF communications and the necessity for robust demodulation techniques.

3.2 Channel Noise in RF reception

In wireless communications, the signal reception is significantly influenced by the propagation environment. The effects of the channel on the transmitted signal can be succinctly represented by the following model [14, 33, 42]:

$$y = h * x + n \tag{1}$$

In Equation 1, y is the received signal, x is the transmitted signal, h represents the channel's impulse response (convolved noise component), and n denotes the additive noise. This model encapsulates how signals are altered during transmission over a wireless medium, focusing on two primary components: the *convolution* with the channel component h and the *additive* noise component n.

1. ***Convolved Noise** Component (h):* The term h encapsulates all transformations the signal undergoes as it propagates from the transmitter to the receiver. This includes i) *Path Loss:* the reduction in signal power intensity as it travels through space, ii) *Fading:* Variations in signal amplitude and phase caused by factors including atmospheric conditions, multipath propogation and object movement, and iii) *Multipath Effects:* The arrival of multiple, time-delayed copies of the signal at the receiving antenna via different paths, leading to inter symbol interference and potential signal distortion.

2. ***Additive Noise** Component (n):* This term encompasses forms of noise and external interferences added to the signal during its transmission, which can originate from both intrinsic sources (like thermal noise from electronic components) and extrinsic sources (such as electromagnetic interference from external environments). This noise affects the quality of the received signal and can introduce errors unless adequately mitigated.

The convolution of x with h describes the combined effect of channel impairments on the signal as it travels from the transmitter to the receiver, often modeled by Rayleigh [31, 36] and Rician [46] distributions. The addition of n models the random disturbances that degrade the clarity and integrity of the communication signal, commonly represented as Additive White Gaussian Noise (AWGN).

Effectively compensating for channel effects requires addressing **both** the additive and convolved noise components comprehensively. Throughout this paper, we explore methods to mitigate these effects, including hardware-induced offsets which are also considered part of the broader spectrum of channel effects.

4 System Design

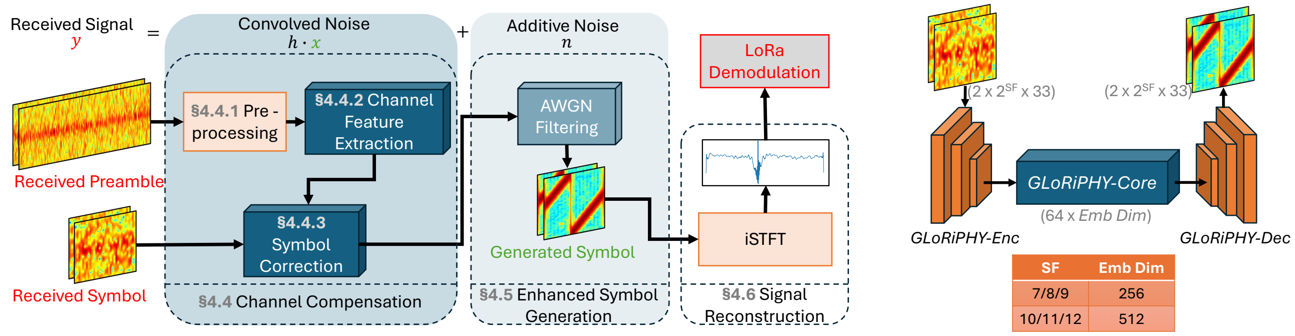
We now present *GLoRiPHY*'s design and implementation details.

4.1 System Model

The primary goal of *GLoRiPHY* is to reconstruct noisy LoRaPHY signals by effectively compensating for the noise and distortions introduced by the environment and hardware during transmission. To achieve this, we come up with several key design choices:

(1) Accurately model and remove both additive and convolved components of the channel noise, thereby facilitating effective *compensation of channel effects*. This capability allows the model to adapt and perform reliably across varying deployment scenarios, eliminating the need to specifically tune the model to noise patterns in new environments.
(2) Develop a compact and scalable model architecture to manage computational efficiency and reduce inference time. This is particularly vital at higher Spreading Factors (SFs), where the exponential increase in frequency domain features demands efficient model design.

Assumptions. We assume that the channel remains stable throughout the duration of a LoRa packet's transmission, a premise supported by real-world studies detailed in Chime [12]. Short-lived



(a) (b)

Figure 2: (a) Overview of *GLoRiPHY* Design. The system independently handles convolved and additive noise components through the *Channel Compensation* and *Enhanced Symbol Generation* phases, respectively, followed by the *Signal Reconstruction* phase that regenerates the signal for LoRa Demodulation. (b) *GLoRiPHY* utilizes *GLoRiPHY-Enc* and *GLoRiPHY-Dec* to manage embedding dimensions processed by *GLoRiPHY-Core*.

active interference that occurs on a time scale shorter than the packet length is outside the scope of our current work.

4.2 System Overview

Figure 2(a) provides an overview of *GLoRiPHY*'s architecture. We delineate the compensation for the convolved and additive components of the channel noise into two independent modules. The *Channel Compensation* module is used to handle the convolved component, while *Enhanced Symbol Generation* addresses the additive component. The *Signal Reconstruction* module reconstructs the signal for LoRa demodulation.

The process begins with the *Channel Compensation* module (Section 4.4). A key insight we make use of in the design of *GLoRiPHY* is the utilization of the received preamble—a sequence containing a known bit pattern—to estimate how the channel has altered the signal and incorporate that knowledge to denoise an incoming LoRaPHY signal. *GLoRiPHY* first pre-processes the received preamble by cross-correlating it with a known reference preamble (Section 4.4.1). Next, the *Channel Feature Extraction* layer (Section 4.4.2), provides the model the ability to interpret and correct channel-induced distortions by extracting relevant information. Subsequently, the *Symbol Correction* layer (Section 4.4.3) utilizes these learned channel features to compensate for the offsets induced by the channel on the received symbol.

It is important to note that cross-correlating the received and expected preamble minimizes the influence of additive noise (as discussed in Equation 1), since additive components are less likely to correlate. This step highlights signal features directly related to the convolved noise, and necessitates a subsequent phase dedicated to addressing Additive White Gaussian Noise (AWGN).

The *Enhanced Symbol Generation* module (Section 4.5) addresses the residual additive noise components. A specialized layer processes the adjusted symbol to filter out white noise, utilizing a generative model to effectively produce a clean STFT of the input symbol. This step cleans the signal by eliminating noise components that may persist after the channel compensation phase.

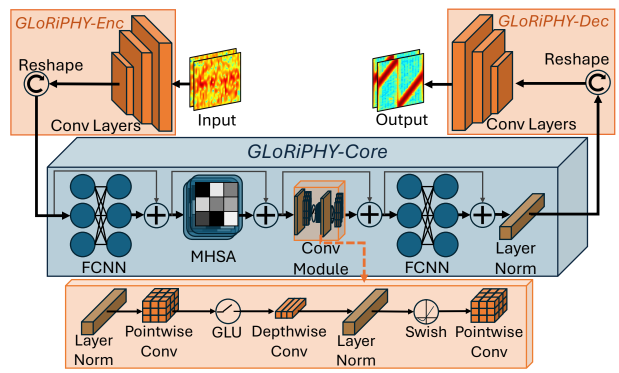


Figure 3: Model Architecture for *GLoRiPHY* Components: *GLoRiPHY-Core*, *GLoRiPHY-Enc* and *GLoRiPHY-Dec*.

Finally, the *Signal Reconstruction* module (Section 4.6) converts the generated STFT back into the time domain using an inverse Short-Time Fourier Transform (iSTFT) operation. The reconstructed signal is suitable for LoRa demodulation, which benefits from the significantly enhanced signal quality.

Next, we detail the machine learning components that underpin the different layers introduced in the aforementioned modules. These components are crucial for the processing capabilities of *GLoRiPHY*, enabling it to effectively learn from and adapt to the channel conditions it encounters.

4.3 *GLoRiPHY* Components

Recent advancements in audio tasks, such as speech enhancement, have highlighted the efficacy of transformer-based neural network architectures [44] in processing spectral inputs and managing long-term temporal features [1, 11, 15, 39]. Meanwhile, Convolutional Neural Networks (CNNs) [21] are recognized for their capability to extract local features. By combining these two architectures, Conformers [15] harness the strengths of both, making them ideal for complex signal processing tasks. Conformers have demonstrated

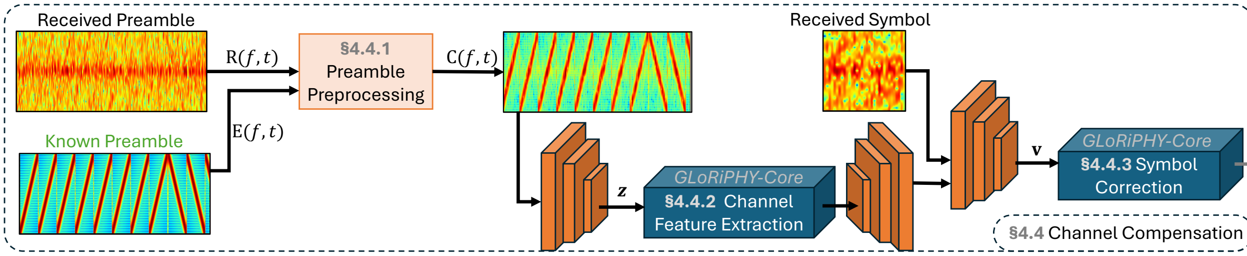


Figure 4: Overview of the *Channel Compensation* Module. The process begins with preprocessing the preamble to estimate the channel impulse response, followed by the Feature Extraction layer learning to identify crucial channel features, and concluding with the Symbol Correction phase that actively compensates for channel effects.

a proven ability to effectively handle various kinds of spectral and temporal feature complexities typical in noisy, real-world audio data. These challenges are analogous to those encountered in LoRa signal processing. Given this alignment, we adopt Conformer as the core learning model for *GLoRiPHY* and refer to it as *GLoRiPHY-Core*.

In Figure 3, we present the model architecture of *GLoRiPHY-Core* in detail. The model integrates two Fully Connected Neural Networks (FCNNs) with a Multi-Head Self Attention (MHSA) [44] and a Convolutional Neural Network (CNN) module. These modules are interconnected with residual connections [17], which facilitate easier training of the deep model by allowing gradients to flow through the network without significant loss.

Transformers typically require input features to be represented in a fixed embedding dimension. In our application, we represent the input as a complex STFT, which ensures a rich feature representation but also expands the feature space along the frequency domain with each increase in Spreading Factor (SF). Specifically, the minimum frequency resolution required to represent a symbol scales as 2^{SF}. Managing such high embedding dimensions could lead to models with billions of parameters, which is impractical for deployment due to substantial resource requirements and long inference times.

To address this challenge, we employ CNN-based encoders and decoders, named *GLoRiPHY-Enc* and *GLoRiPHY-Dec*, respectively. As illustrated in Figure 3, *GLoRiPHY-Enc* and *GLoRiPHY-Dec* each consist of a series of CNN layers, coupled with a reshape operation, designed to effectively transform the high-dimensional STFT data into the specified embedding dimension and vice versa. These components efficiently compress the high-dimensional inputs into compact latent representations suitable for processing by *GLoRiPHY-Core*. As illustrated in Figure 2(b), these components work in conjunction to optimize both learning efficiency and computational tractability, ensuring *GLoRiPHY* is practically deployable.

4.4 Channel Compensation

Next, we explain how *GLoRiPHY* compensates for the *convolved component* of the channel noise. Figure 4 illustrates the working of this module.

The **key idea** behind the channel compensation module is to utilize the known preamble transmitted at the start of each LoRaPHY packet as a reference for determining the channel effects.

4.4.1 Preamble Preprocessing. Characterizing the channel's effect on the preamble involves addressing several challenges. Transmissions over long distances severely attenuate the preamble, introducing a blend of convolved and additive noise components (Equation 1). Isolating the effects of one in the presence of the other becomes particularly challenging. The convolved noise components manifest as frequency/time offsets induced by factors including inaccurate hardware oscillators and strong reflecting surfaces causing multipath fading. To effectively identify such offsets, we estimate the Channel Impulse Response (CIR) by cross-correlating the received preamble with the expected preamble sequence.

The cross-correlation is performed in the time-frequency domain to efficiently capture features that characterize the channel's modifications to the signal in terms of time delay, amplitude attenuation, and phase shifts.

Recall Equation 1, where the incoming signal is represented as $y = h * x + n$. In the frequency domain, the convolution operation ($*$) translates to multiplication, making the equation simpler to handle mathematically. Ignoring the additive noise component, the equation in the frequency domain becomes:

$$Y = H \cdot X$$

To estimate the channel characteristics (H), we multiply both sides of the equation by the complex conjugate of X (X^*), which yields:

$$Y \cdot X^* = H \cdot X \cdot X^* \quad \Rightarrow \quad H = Y \cdot X^*$$

This operation effectively estimates H, the channel's frequency response, by utilizing the properties of complex conjugates. In *GLoRiPHY*, the known preamble (or its complex conjugate) helps in estimating the channel in the frequency-time domain.

Mathematically, this process is summarized as follows:

1) STFT Computation: Let $r(t)$ represent the received preamble, and $e(t)$ the expected preamble signal. Let the Short-Time Fourier Transform (STFT) of the received and expected preambles be:

$$R(f,t) = \mathrm{STFT}\{r(t)\}, \quad E(f,t) = \mathrm{STFT}\{e(t)\}$$

where the STFT function is defined as:

$$\mathrm{STFT}\{x(t)\} = \int x(t) w(t-\tau) e^{-j2\pi f\tau}\, d\tau$$

Here, $w(t)$ represents the windowing function, which is typically a Hamming window in our application.

2) Cross-Correlation: To capture the channel effects, we compute the cross-correlation of the computed STFTs:

$$C(f,t) = R(f,t) \cdot E^*(f,t)$$

where $E^*(f,t)$ is the complex conjugate of $E(f,t)$.

As demonstrated in Figure 4, cross-correlating with a known preamble allows us to form a feature representation that largely ignores extraneous noise components, such as thermal noise, and instead focuses on how the signal of interest has been modified by the channel. This time-frequency domain representation is particularly effective in capturing the impact of hardware offsets and frequency-selective effects like fading and interference, even in highly noisy scenarios. These features are crucial as they significantly impact the quality and reliability of the received signal and are expected to affect the rest of the packet similarly. However, given the complexity and richness of the channel features, it becomes essential to incorporate a learnable layer within the system to efficiently extract and leverage the relevant features.

4.4.2 Channel Feature Extraction. To effectively utilize the Channel Impulse Response (CIR), accurate extraction of pertinent features is crucial. For this purpose, we employ the *GLoRiPHY-Enc* layer to learn a compact latent representation, efficiently encoding the high-dimensional complex CIR into a feature vector **z**. This encoded representation **z** captures the essential channel characteristics while significantly reducing the computational overhead required for subsequent processing steps.

The representation **z** is then fed into the key learning layer, *GLoRiPHY-Core*, where the integration of the self-attention and convolutional modules allows for comprehensive feature extraction. *GLoRiPHY-Core* effectively captures both local and global dependencies within the encoded feature space. Following feature extraction, the output from *GLoRiPHY-Core* is upscaled back to the dimensions of a single symbol input using *GLoRiPHY-Dec*. This upscaling process is meticulously designed to ensure that each value in the complex Short-Time Fourier Transform (STFT) of the received symbol is precisely mapped, reflecting how each component has been influenced by the channel.

4.4.3 Symbol Correction. In the final phase of the *Channel Compensation* module, we employ another *GLoRiPHY-Core* layer, to compensate for the learned channel effects on the received payload symbol. Initially, a *GLoRiPHY-Enc* layer encodes both the learned features—which represent the channel effects—and the complex STFT of the received symbol into a combined feature vector, **v**.

The encoded feature vector **v** captures both the symbol that needs to be denoised and the extracted features indicative of the channel induced effects. Thus, the *GLoRiPHY-Core* layer processes this composite feature vector, adapting its approach to actively compensate the channel distortions. This layer adjusts the features in an attempt to effectively restore the original signal characteristics.

The machine learning layers employed in the Channel Feature Extraction and Symbol Correction stages are crucial for the effective operation of *GLoRiPHY*. These layers require training on datasets that represent real-world scenarios, enabling the models to learn to extract and compensate for various channel effects. We explain the training methodology in Section 5.

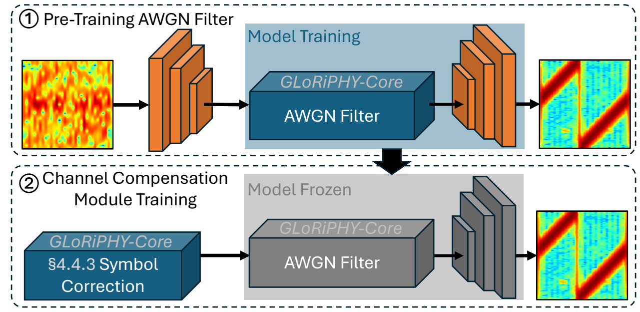


Figure 5: 1) Pre-training of AWGN Filter on AWGN-Perturbed Data, and 2) Its Utilization in Guiding *Channel Compensation* Module Training.

4.5 Enhanced Symbol Generation

After the symbol has been processed by the *Channel Compensation* module, the residual additive white noise still remains. To address this challenge, we introduce an Additive White Gaussian Noise (AWGN) filter.

Key Idea: Given that the additive component of noise is assumed to be random, it does not need to be 'learned' from the preamble. Instead, we use a pre-trained AWGN filter designed to clean symbols perturbed by AWGN.

Pre-training AWGN Filter: The AWGN filter is expected to receive symbols compensated for convolved noise, from the preceding module, leaving residual additive noise. For training, we simulate such symbols that the filter would typically encounter. This is achieved by taking clean LoRaPHY symbols and adding White Gaussian Noise at various Signal-to-Noise Ratio (SNR) levels to both the phase and amplitude of the symbol's STFT.

Filter Design: Following the approach outlined in Figure 2(b), we employ *GLoRiPHY-Enc* to encode the noise-perturbed STFT input, which is then processed by a *GLoRiPHY-Core* layer. The output from *GLoRiPHY-Core* is upscaled by *GLoRiPHY-Dec* to regenerate the cleaned STFT. Figure 5 visually depicts the filter design.

Details on the loss function and the overall model training methodology are discussed in Section 5.

4.6 Signal Reconstruction

To ensure that the output signal is in a format usable for standard LoRa demodulation processes, we introduce the *Signal Reconstruction* module. While *GLoRiPHY* generates a symbol STFT compensated for the channel effects, LoRa systems require the signal in the time domain for demodulation. Consequently, the generated STFT must be converted back to the time domain.

This conversion is achieved through the Inverse Short-Time Fourier Transform (iSTFT). The iSTFT operation mathematically reverses the STFT to reconstruct the time-domain signal, which can be represented by:

$$x(t) = \text{iSTFT}\{X(f,\tau)\}$$

where the iSTFT function is defined as:

$$\text{iSTFT}\{X(f,\tau)\} = \sum_{\tau}\sum_{f} X(f,\tau)w(t-\tau)e^{j2\pi f(t-\tau)}$$

Here, $X(f,\tau)$ denotes the STFT coefficients, $w(t)$ represents the window function used during the STFT, and t indexes time.

5 *GLoRiPHY* Implementation

We now describe the steps we take to train the models in *GLoRiPHY*.

Weighted Dual Loss Function: *GLoRiPHY* employs a dual component loss function designed to achieve two primary objectives that ensure: 1) the model generates an accurate symbol STFT with correct phase and amplitude features across frequency and time; 2) the reconstructed signal from the STFT is directly usable for LoRa demodulation.

The first objective is addressed by computing an element-wise Mean Square Error (MSE) between the generated symbol STFT, $\mathcal{S}_{\text{Gen}}$, and the ground-truth symbol STFT, $\mathcal{S}_{\text{GT}}$. The second objective is achieved by assessing the categorical cross-entropy loss between the demodulation result on the reconstructed signal, $\mathcal{D}_{\text{Gen}}$, and the actual symbol ID, $\mathcal{D}_{\text{GT}}$. The combined loss function is then formulated as:

$$\text{Loss} = \alpha \cdot \text{MSE}(\mathcal{S}_{\text{Gen}}, \mathcal{S}_{\text{GT}}) + \beta \cdot \text{Cross-Entropy}(\mathcal{D}_{\text{Gen}}, \mathcal{D}_{\text{GT}})$$

Here, α and β are weights that balance the emphasis on accuracy of STFT reconstruction versus demodulation efficacy. We assign a higher weight to the MSE component to emphasize signal fidelity across frequency, time, amplitude, and phase.

Curriculum Learning: The training regimen adopts a curriculum learning approach [3]. This approach begins with the model learning to differentiate between symbols under minimal noise conditions. This phase helps the model to capture essential features and their interrelationships. Then, training data complexity is gradually increased by introducing higher noise levels, thus enhancing the model's robustness and reliability under varied and challenging conditions. This step-wise escalation in training complexity is a proven strategy for complex learning tasks and has been shown to improve learning efficiency [16].

Channel Compensation Module Training: As depicted in Figure 5 and discussed in Section 4.5, we first train the AWGN filter independently. Then, the filter's weights are frozen and used to guide the training of the *Channel Compensation* module.

The **key intuition** underpinning our training strategy is that once the AWGN filter is trained and its weights are frozen, it becomes tuned to correct only the types of noise it was trained on, specifically, symbols perturbed with additive noise. In essence, the pre-trained AWGN filter acts as a strict criterion during the training of the channel compensation module: it can only reduce the loss function when it receives an input that aligns with its training, i.e., symbols that are offset-free and contain **only** additive noise. This strategic setup ensures that the channel compensation module is incentivized to focus on compensating for the more complex convolved noise components first, leaving only residual additive noise in the output.

Such a configuration is crucial because it effectively isolates the training processes for the two distinct types of noise components—convolved and additive. By doing so, it streamlines the overall model training, allowing each module to specialize and excel in correcting specific noise types without interference from the other.

Compact Embedding Representation: As introduced in Section 4.4, *GLoRiPHY* utilizes *GLoRiPHY-Enc* and *GLoRiPHY-Dec* to encode and decode high-dimensional inputs into a compact embedding representation suitable for *GLoRiPHY-Core*'s processing. We use embedding dimension of 256 for SF 7, 8, and 9. For higher Spreading Factors (SF 10, 11, and 12), we increase the embedding dimension to 512. This adjustment helps maintain a balance between model size and accuracy.

Moreover, the embedding dimension is designed to be tunable, allowing for adjustments based on specific deployment scenarios. This flexibility ensures that *GLoRiPHY* can be optimized for performance or resource usage as required, making it adaptable to a wide range of operational environments and application needs.

***GLoRiPHY* Deployment:** *GLoRiPHY* must be pre-trained on data that reflects real-world channel conditions to effectively learn to mitigate channel effects. Once trained, the model can either be directly installed on gateways, provided there are adequate computational resources, or hosted as a cloud service to take advantage of greater computing power.

6 Evaluation

This section details the experimental framework designed to evaluate *GLoRiPHY* using both real-world and simulated datasets.

6.1 Evaluation Setup

6.1.1 Datasets. Below, we elaborate on our datasets.

1. Real-World Dataset. Our real-world test setup utilizes SX1272-based Fipy microcontrollers [32] to transmit LoRa packets with random payloads. The packets are received using a USRP B210, with the GNURadio implementation for LoRaPHY reception on SDRs [48], adapted to facilitate the data collection process. The network operates at a center frequency of 868 MHz with a bandwidth of 125 kHz and a Spreading Factor (SF) of 8, received at a sampling rate of 1 MHz. As depicted in Figure 6, our data collection spans a diverse range of environments to robustly train and test *GLoRiPHY* under varied conditions. Below, we explain our four chosen experimental environments in detail.

(a) Indoor: Over 100k packets are collected from 10 nodes across an *indoor* area of approximately 14,300 square meters, over 10 days. The devices are strategically placed on different floors across three buildings to introduce variable signal attenuation and multipath effects. We also lower transmit power settings to simulate longer-distance signal attenuation. Two USRP instances were used to capture a range of hardware offsets within the dataset.

(b) Semi-Outdoor: Three nodes are deployed on different levels between two buildings separated by vegetation, walkways, a road, and open carparks, offering a *semi-outdoor* scenario.

(c) Outdoor: Nodes are placed in a community park with heavy vegetation and undulating terrain, adjacent to a busy road and high-rise buildings, encompassing a challenging *outdoor* setting. The

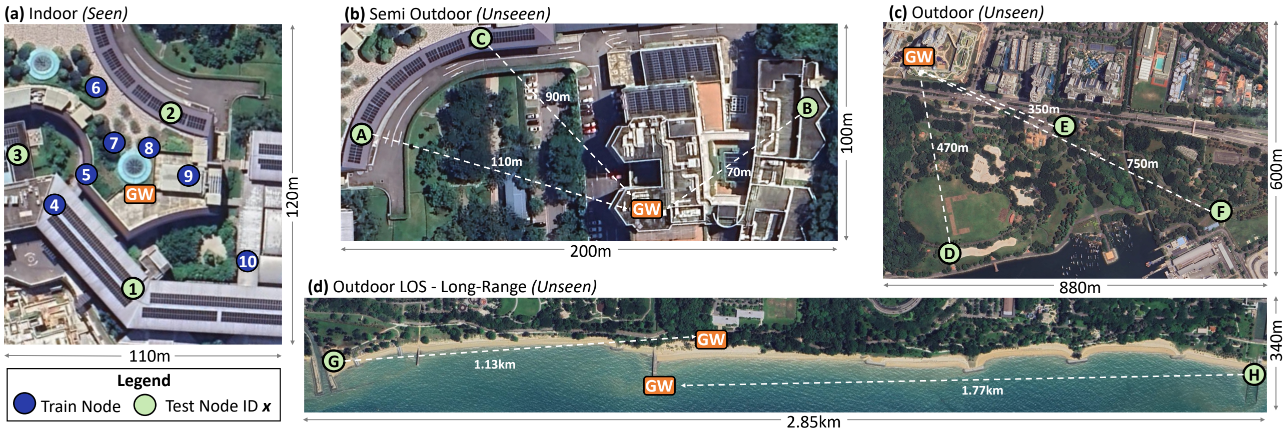
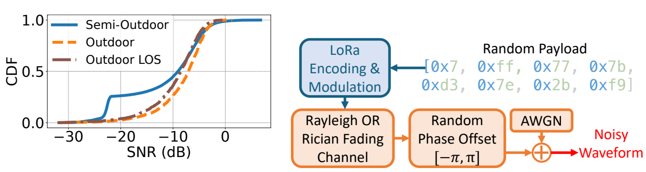


Figure 6: Real-world Deployment—Illustrating a diverse range of testing environments to evaluate *GLoRiPHY*'s adaptability: (a) *Indoor:* Nodes placed on different levels between neighboring buildings (Seen), (b) *Semi-Outdoor:* Nodes positioned in buildings separated by vegetation, walkways, and carparks (Unseen), (c) *Outdoor:* Nodes located in a community park surrounded by high-rise buildings (Unseen), (d) *Outdoor LOS (Long-Range):* Transmission covering distances up to 1.77 km (Unseen).



(a) (b)

Figure 7: (a) Cumulative Distribution Function (CDF) of SNR observed in real-world *unseen* environments, (b) Flowchart Representing the Process of Simulated Data Generation.

park's dynamic environment, including significant pedestrian and car movement, introduces additional variables.

(d) Outdoor LOS (Long-Range): For long-range communication tests, nodes are set up along the shoreline. One node transmits over a distance of 1 km with partial obstructions, and another over open water with clear line of sight at 1.77 km.

We use only the *indoor* dataset to train *GLoRiPHY*, and therefore refer to it as the *seen* environment, while the remaining environments are considered *unseen* by the model. For testing purposes, we utilise all four datasets, and focus only on packets that exhibit at least one symbol error. In total, we collected over 38,000 packets that meet this criterion, with each test node contributing at least 350 packets. The unseen dataset was collected using a new USRP device not used to collect the training dataset, to ensure that the model's performance is not biased by familiar hardware characteristics. In Figure 7(a), we present the cumulative distribution function (CDF) to illustrate the distribution of the Signal-to-Noise Ratio (SNR) across our unseen environments. Overall, the SNR ranges from −32.1dB to 6.7dB with a mean SNR of −11.6dB. This rigorous testing framework is designed to thoroughly evaluate *GLoRiPHY*'s robustness and adaptability.

2. Simulated Dataset. While the real-world dataset allows us to evaluate *GLoRiPHY*'s performance in practical settings, we utilise simulated data to test the model's generalization across different SFs and a broader range of multi-path environments. For this, we simulate LoRaPHY data using MATLAB's Communication Toolbox [41]. The dataset consists of LoRaPHY packets with random payloads, which are perturbed using Rayleigh/Rician fading channels to simulate the convolved component of channel noise. Additive White Gaussian Noise (AWGN) is added to mimic the additive noise component. The simulation process is summarised in Figure 7(b). Note that while we introduce random phase offsets to our simulated data, we do not specifically simulate hardware-induced offsets such as Carrier Frequency Offset (CFO) and Sampling Frequency Offset (SFO). This allows us to focus on the impact of channel fading.

Simulation parameters are chosen to reflect real-world scenarios closely. The channel parameters are randomly varied across a wide range, representing a diverse mix of urban, rural, indoor, and outdoor scenarios, to robustly test the model under various noise conditions. Variations include the number of paths, average time delays and attenuation faced by the multipath components, and the Doppler frequency for Rayleigh fading. Additionally, the K-factor is varied for Rician fading, to reflect different levels of line-of-sight components. The Signal-to-Noise Ratio (SNR) of the simulated data is predominantly uniformly distributed in the range $[-25, 0]$dB. Overall, data is perturbed with 1,000 distinct simulated channels.

6.1.2 Baseline. We benchmark *GLoRiPHY* against traditional LoRaPHY demodulation and the state-of-the-art ML-based demodulation solution, NELoRa. The traditional LoRaPHY demodulation process is implemented as detailed in [48]. NELoRa's model, per the open-sourced implementation, is trained initially with AWGN-perturbed data, then fine-tuned with the specific evaluation scenario's dataset. Notably, extending the training iterations improved performance. Hence, we present the results with several additional training cycles compared to the original recommendations.

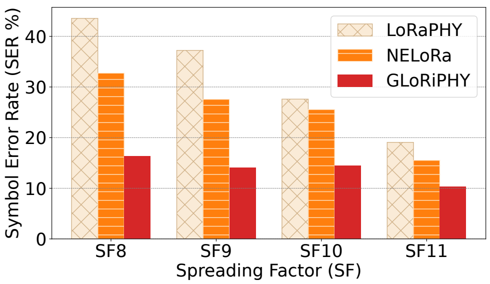


Figure 8: Evaluation of Simulated Data to Assess the Impact of Multipath Effects on Symbol Error Rate (SER) Across Different Spreading Factors (SFs).

6.1.3 *Evaluation Metrics.*

(1) **Symbol Error Rate (SER %)**: Measures the percentage of LoRaPHY symbols *incorrectly* demodulated. A lower SER signifies a more effective demodulation solution. We evaluate the SER under varying conditions to assess solution robustness across different noise environments and transmission scenarios. We report the average SER across all tested packets for each configuration.
(2) **Inference Time**: The time required to identify the symbol upon reception, with a lower time enhancing system responsiveness and practical viability.

Given our focus on a diverse range of fading channels, we do not consider SNR gain as a primary evaluation metric. Theoretical and empirical studies have suggested that at equivalent SNR levels, different fading channels can yield varying SERs [7, 37]. For instance, under similar SNR conditions, an AWGN channel might exhibit a lower SER than a Rayleigh fading channel. This implies that in environments characterized by diverse multipath fading, SNR alone does not adequately reflect the channel conditions or the effectiveness of a demodulation solution. Hence, our evaluation prioritizes SER metric that directly measure the impact on communication performance, offering a more accurate reflection of the system's capability in practical deployment scenarios.

6.1.4 *Evaluation Scenarios.* To comprehensively evaluate *GLoRiPHY* across different configurations, we explore a series of Research Questions (RQs) designed to test various aspects of the system. This structured evaluation aims to provide insights into both the efficacy and practicality of our model under diverse conditions.

In Section 6.2, we compare the SER in both simulated and real-world environments to assess *GLoRiPHY*'s effectiveness. In Section 6.3, we examine practical deployment aspects of *GLoRiPHY*, particularly model size and inference time. Finally, In Section 6.4, we conduct an ablation study to validate our design decisions.

6.2 Overall Performance

6.2.1 *RQ1: How effective is GLoRiPHY at compensating for multipath induced channel effects?*

Setup. We utilize the simulated dataset to evaluate *GLoRiPHY*'s ability to compensate for multipath noise since we are able to better control and vary the amount of multi-path effect in the data. This dataset simulates a large number of unique multipath channels. We train *GLoRiPHY* using 80% (800) of the simulated channel instances, reserving the remaining 20% for testing. This partitioning helps evaluate the ***generalization capability*** of *GLoRiPHY* on unseen channel instances. We also tune the NELoRa model on the same dataset for a fair comparison, repeating this process for Spreading Factors (SFs) 8 through 11.
Results. As summarized in Figure 8, *GLoRiPHY* maintains a low SER over all evaluated Spreading Factors, ranging from a minimum of 10.3% to a maximum of 16.3%. Notably, the SER for traditional LoRa Demodulation is as high as 43.75% at SF8, decreasing with higher SFs. This observation is in line with the findings on the increased resilience of LoRaPHY to multi-path effects at higher SFs [38]. *GLoRiPHY* consistently outperforms LoRaPHY across all SFs, demonstrating its effective compensation for multipath effects.

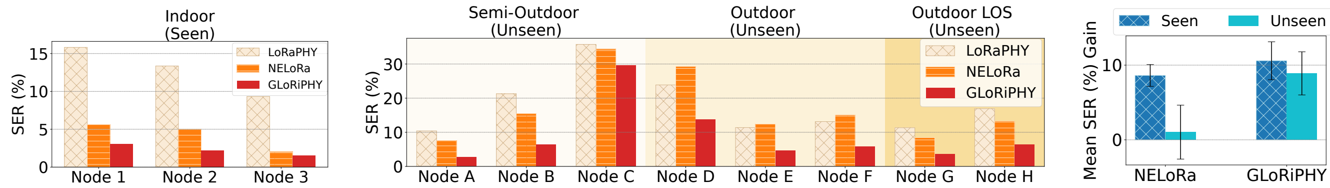
NELoRa, although trained on the same dataset, performs worse than *GLoRiPHY*, largely due to its channel-agnostic design. While NELoRa does improve upon standard LoRa demodulation, the impact of multipath noise is still significant, with the SER reaching 32.73% at SF8. In contrast, *GLoRiPHY* demonstrates far superior noise mitigation, achieving SER levels at SF8 that are comparable to NELoRa's performance at SF11. This significant difference underlines *GLoRiPHY*'s capability to work with non AGWN noise.

6.2.2 *RQ2: Can GLoRiPHY compensate for hardware offsets and real-world noise?*

Setup. This scenario tests *GLoRiPHY*'s effectiveness in correcting hardware-induced offsets and channel effects using the real-world *Indoor* dataset consisting of 10 nodes. We evaluate the model's performance on three distinct nodes (node #1, #2 and #3), that are well separated physically and unseen in the training model (see Figure 6). Using a leave-one-out approach, data from all other nodes except the test node are used for training.
Results. The evaluation results are shown in Figure 9(a). Consistent with our findings from the simulated dataset, *GLoRiPHY* maintains low SERs across all test nodes. Notably, unlike other baselines, hardware offsets are *not* specifically corrected for in *GLoRiPHY*'s data processing. Instead, *GLoRiPHY* corrects for such offsets as well using the preamble context. The low SER rates on unseen nodes demonstrate *GLoRiPHY*'s robust capability to compensate for both wireless channel effects and hardware-induced errors.

6.2.3 *RQ3: Does GLoRiPHY generalise to unseen real-world environments?*

Setup. We assess the generalizability of *GLoRiPHY* in three distinct real-world environments: *Semi-Outdoor*, *Outdoor*, and *Outdoor-LOS*, testing with distances up to 1.77 km, as introduced in Section 6.1.1. The data is collected using a new USRP, not seen during the model's training. The models trained for the previous real-world experiment are used for testing, without further tuning.
Results. As illustrated in Figure 9(b), receptions from the eight test nodes exhibit SERs ranging from 10% – 35% when using standard LoRaPHY. Notably, despite *GLoRiPHY* not being trained with data from these environments, it consistently outperforms the considered baselines by maintaining the lowest SER, underscoring its robust generalization capabilities due to its channel-aware design.



(a) (b) (c)

Figure 9: (a) Comparison of Symbol Error Rate (SER) in Real-World *Seen* Environment, (b) Comparison of Symbol Error Rate (SER) in Real-World *Unseen* Environments, (c) Average Symbol Error Rate (SER) Gain over LoRaPHY in an *Unseen* Environment compared to an Environment *Seen* by the model in its training set.

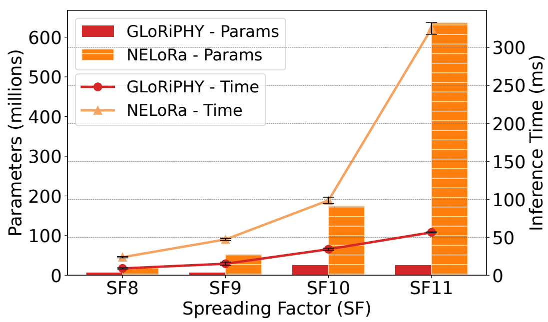


Figure 10: Comparison of Model Size and Inference Time Between *GLoRiPHY* and NELoRa.

To quantify the performance difference between NELoRa and *GLoRiPHY* in varying environments, we show the relative improvement over LoRaPHY in the two environments: (1) similar to the training data (Section 6.2.2), and (2) the unseen environments mentioned above. As shown in Figure 9(c), *GLoRiPHY*'s SER improvement over LoRaPHY drops by about 1.7%, from 10.59% to 8.9%, when the environment changes to the unseen case. On the other hand, the performance drop for NELoRa is much more significant, from 8.62% to 1.03%, indicating low adaptability to new conditions.

It should be noted that the relative order in performance among LoRaPHY, NELoRa and *GLoRiPHY* remains consistent in the *Semi-Outdoor* and *Outdoor-LOS* environments. However, in the *Outdoor* environment, we note that NELoRa's performance drops below LoRaPHY. This can be attributed to the challenging conditions in this environment, which features heavy multipath and attenuation from surrounding high-rise buildings, undulating terrain, and dense vegetation—factors that are starkly different from those in the training (*Indoor*) environment.

6.3 Runtime Performance

In this section, we address the following question: *What is the overhead in utilizing GLoRiPHY to denoise an incoming signal?*
Setup. We evaluate the effect of model size (in terms of number of parameters) on inference time with various Spreading Factor (SF) using NELoRa as the baseline. In the evaluation, we process a batch of 64 symbols and determine the time it takes for the model to identify the symbol. Both models are tested under identical conditions on an NVIDIA A6000 GPU paired with an Intel Xeon CPU (clock frequency of 2.6 GHz). Each experiment is performed 100 times and we report the mean, and standard deviation of observed values.
Results. As presented in Figure 10, *GLoRiPHY* maintains a compact model size. Our use of *GLoRiPHY-Enc* and *GLoRiPHY-Dec* to encode and decode high-dimensional features allows us to keep the model size much smaller compared to NELoRa. As we maintain a small embedding size for *GLoRiPHY-Core*, we observe the model size increases only slightly for SFs 10 and 11, where we use a higher embedding size of 512, in comparison to 256 for SFs 8 and 9. In contrast, NELoRa 's model size increases exponentially with SF. We note that at SF12, NELoRa's model increases to 2.4 billion parameters, while the model size of *GLoRiPHY* is less by almost two orders of magnitude at 25.5 million parameters. In terms of memory footprint, *GLoRiPHY* occupies 28.78 MB, 100.91 MB and 389.32 MB on the disk for models with an embedding dimension of 256, 512 and 1024 respectively.

Moreover, while compute resource requirements remain a concern, the increasing model size for NELoRa also leads to higher inference time that increases exponentially with SF. In contrast, *GLoRiPHY* experiences only a gradual increase in inference time, further underscoring its efficiency and practicality for deployment in real-world applications.

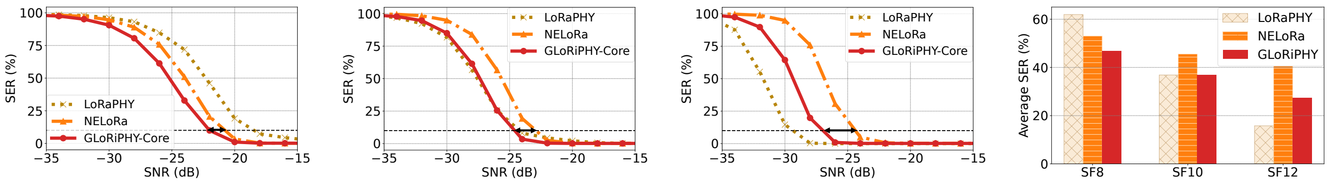
We highlight that *GLoRiPHY* offers a tunable approach to select the embedding size, providing flexibility in adapting to various computational resource environments and latency requirements. We explore this capability further in Section 6.4.3.

6.4 Ablation Study

In this section, we explore the following question: *How do the different modules introduced in GLoRiPHY contribute to its overall effectiveness?*

6.4.1 Channel Compensation Module (§ 4.4).

Setup. To assess the significance of the *Channel Compensation* module in providing *GLoRiPHY* its channel awareness capabilities, we conduct a series of tests using the simulated dataset described in Section 6.1.1. We first deactivate the entire channel compensation module, training the system solely with the AWGN filter on offset-corrected data. Subsequently, we reintegrate the channel



(a) (b) (c) (d)

Figure 11: Symbol Error Rate (SER) vs. Signal-to-Noise Ratio (SNR) Comparing *GLoRiPHY*'s AWGN Filter, NELoRa and LoRaPHY on AWGN-Perturbed Data for (a) SF 8 (b) SF 10 and, (c) SF 12. (d) Results comparing average SERs of different SFs over SNR range of [-35,-15]db.

Test Scenario	SER (%)
Baseline - NELoRa	32.73
w/o Channel Compensation Module (§ 4.4)	31.21
w/o Preamble Preprocessing (§ 4.4.1)	17.81
GLoRiPHY	**16.3**

Table 1: Ablation Study Results for the Channel Compensation Module.

compensation but exclude one of its components, the *Preamble Preprocessing* stage (§ 4.4.1). Instead, the raw preamble is used as input to train the Channel Feature Extraction layer. These tests are conducted at SF8, which is observed to be the most challenging configuration in our earlier evaluations.

Results. The results summarized in Table 1 reveal the critical role of the channel compensation module. Removing the module entirely results in a substantial increase in SER from 16.3% to 31.21%, indicating a significant degradation in performance compared to when *GLoRiPHY* is fully enabled.

The impact of the preamble preprocessing yields a much less notable degradation, with SER increasing from 16.3% to 17.81%. This finding underscores that while preamble preprocessing is helpful in suppressing unwanted noise components and assisting in the extraction of crucial features from the preamble, its impact is clearly much less significant compared to the machine learning based channel feature extraction module. Therefore, *GLoRiPHY* would still perform reasonably well if the raw preambles are used directly as input to the Channel Feature Extraction layer.

The evaluation validates our design decision to integrate the channel compensation module into *GLoRiPHY*. It demonstrates the module's pivotal role in enabling *GLoRiPHY* to accurately account for and counteract complex channel effects.

6.4.2 Enhanced Symbol Generation (§ 4.5).

Setup. In this segment of our ablation study, we examine the AWGN filter within the *Enhanced Symbol Generation* module to assess its effectiveness in filtering out Additive White Gaussian Noise (AWGN). For this evaluation, we employ a simulation framework similar to the one introduced in Figure 7(b), while omitting the multi-path noise component (Rayleigh/Rician Fading Channel). The AWGN is added in accordance with the procedures outlined by NELoRa-Bench [9] for a controlled evaluation. Both our AWGN Filter and NELoRa are trained and tested on this data to compare their performance.

Results. Figure 11 presents the results, comparing the Symbol Error Rate (SER) achieved by *GLoRiPHY*'s AWGN filter and NELoRa at various Signal-to-Noise Ratio (SNR) values, for SF 8, 10, and 12. *GLoRiPHY*'s AWGN filter outperforms NELoRa across all configurations, maintaining a significantly lower SER. The SNR gain of *GLoRiPHY* over NELoRa, at a SER of 10%, is 1.25 dB, 1.87 dB and 2.5 dB for SF8, 10 and 12 respectively.

As noted in our earlier evaluations (Section 6.3), *GLoRiPHY* maintains a smaller model size and faster inference times compared to NELoRa. Importantly, while NELoRa's model remains the same in size and complexity, the AWGN filter component of *GLoRiPHY* uses only a fraction of the resources reported for the overall sytstem, both in terms of model parameters and inference times. Specifically, for the SF 12 test, NELoRa's model size escalates to 2.4 billion parameters, whereas *GLoRiPHY*'s AWGN filter operates with only 12.7 million parameters, underscoring its efficiency and effectiveness.

However, in contrast to our findings with scenarios involving multi-path noise (Section 6.2.1), we observe that as SF increases, the SER gain achieved by LoRaPHY begins to surpass that of the current AI-based solutions when faced with AWGN alone. As summarised in Figure 11(d), the rate of SER reduction by LoRaPHY exceeds that of both NELoRa and *GLoRiPHY* at SF 12. This observation can be attributed to LoRaPHY's improved ability to handle random noise at higher SFs, as the demodulation process utilizes an FFT over larger time frames, averaging out noise over a longer period. The current AI-based solutions relying on STFT inputs do not benefit similarly. We elaborate on this in the discussion (Section 7).

6.4.3 Embedding Size.

Setup. As previously discussed in Section 6.3, *GLoRiPHY* achieves significantly lower inference times and smaller model sizes across various Spreading Factors (SFs) compared to baseline models. This efficiency is largely due to how *GLoRiPHY-Enc* and *GLoRiPHY-Dec* manage to represent high-dimensional features in compact latent embedding dimensions. Moreover, these components offer flexibility to adjust the embedding dimension based on deployment scenarios. Having demonstrated performance gains with a prudently chosen embedding dimension of 512 for SF 10, we now explore the impact of varying this dimension. We extend our evaluation of the

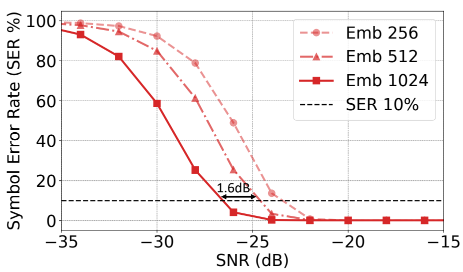


Figure 12: Symbol Error Rate (SER) vs. Signal-to-Noise Ratio (SNR) trends on varying *GLoRiPHY*'s Embedding Dimension.

AWGN filter from the previous section by comparing performance using latent representations of 256 and 1024 dimensions.

Results. Figure 12 illustrates the effect of varying the embedding dimension on the model's learning ability. In previous tests with an embedding dimension of 512 for SF 10, we observed an SNR gain of 1.87 dB over NELoRa at 10% SER. Increasing the embedding dimension to 1024, while raising the model size to 50.5 million parameters, provides an *additional* SNR gain of 1.6 dB. This enhancement also demonstrates a marked improvement compared to standard LoRaPHY at SF 10, contrasting our earlier findings with a lower embedding dimension of 512, as depicted in Figure 11(b).

Conversely, reducing the embedding dimension to 256 results in a much smaller model, with only 3.2 million parameters, but at the cost of degraded performance. This evaluation confirms that while higher embedding dimensions can significantly enhance performance by enabling more detailed feature representation, they also increase the model's size and computational requirements. Conversely, lower dimensions offer efficiency gains but may compromise model's effectiveness. This flexibility allows *GLoRiPHY* to be tailored to specific needs and deployment constraints, balancing performance with computational resource availability.

7 Discussion

In this section, we discuss *GLoRiPHY*'s generalizability, deployment considerations, limitations as well as opportunities for future work.

LoRa's Senstivity to Multi-path Fading: Our analysis of the simulated dataset reveals that introducing multi-path fading can significantly increase the SER at a given SNR, compared to the scenarios where only AWGN is present. This observation is consistent with previous theoretical studies [7], which demonstrate that even a single additional signal path can substantially degrade SER in comparison to an ideal single-path scenario. These findings underscore the critical impact of multi-path conditions on LoRa's performance and highlight the importance of effective mitigation strategies to enhance system robustness.

Broader Applicability: The methodology introduced by *GLoRiPHY* is not confined to LoRa technologies alone. Its underlying principles can be adapted to other narrowband communication technologies that utilize a known preamble, broadening its scope.

Integration with FEC: It may be possible to incorporate *GLoRiPHY* as a replacement for traditional Forward Error Correction (FEC) strategies, which rely on redundancies to correct errors during decoding. By injecting known pilot symbols into the packet, *GLoRiPHY* could clean the signal more effectively upfront. This approach could redefine FEC implementation, shifting some redundancy functions from decoding to demodulation, where *GLoRiPHY* can actively correct errors based on the channel context rather than relying solely on redundancies. This could lead to more efficient use of bandwidth and reduced power consumption, as the system would need to handle fewer redundant bits.

Importance of Diverse Training Data: While we have demonstrated *GLoRiPHY*'s generalization capability in urban multi-path scenarios, its performance can be further enhanced with exposure to nuanced channel characteristics found in environments such as rural areas with distinct signal attenuation characteristics, train tunnels exhibiting higher Doppler frequencies, amongst other deployment scenarios. Utilizing such diverse training data not only improves the model's robustness but also its generalizability. As *GLoRiPHY* encounters a broader spectrum of channel conditions, its capacity to adapt and correct unfamiliar distortions increases, potentially reducing the error rates even further.

***GLoRiPHY* Training Overhead:** Training a *GLoRiPHY* model tested in Section 6.2.2 from scratch requires approximately 57 GPU hours. This includes around 30 hours to train the AWGN filter, followed by the remaining hours for *Channel Compensation* module training. However, given the generalisation capability of *GLoRiPHY*, this is a one time effort. Once trained, the model can be deployed across a diverse set of environments without the need for retraining.

Limitation of Simulated Dataset Used: While the simulated dataset employed in our study provides a controlled environment to evaluate the performance of *GLoRiPHY*, it does not include hardware-induced offsets such as Carrier Frequency Offset (CFO) and Timing Offset (TO). Future studies could enhance the realism of the simulated data by incorporating these hardware offsets into the simulation framework. Alternatively, adopting an emulation approach that utilizes clean chirps collected from real-world environments could also help bridge this gap.

Future Work: Our evaluations reveal that for offset-free data, under conditions of completely random noise, LoRaPHY demonstrates comparable or even improved performance at Spreading Factors (SF) ≥ 10, compared to *GLoRiPHY*'s AWGN filter and NELoRa. While increasing the model's scale, particularly the embedding dimension, has shown potential for enhancing performance (see Figure 12). We believe there is scope for future work in this area, including refining input preparation strategies and training methodologies.

8 Conclusion

We present *GLoRiPHY*, a generative framework designed to enhance the reception quality of LoRaPHY signals through a channel-aware denoising mechanism. Central to *GLoRiPHY* is the use of the known preamble of LoRaPHY signals to discern how the propagation channel modifies the received signal. This channel-aware capability allows *GLoRiPHY* to enable performance improvements in long-range communication, including those in unseen environments. Additionally, *GLoRiPHY* incorporates a tunable approach to manage a compact model size, enhancing its flexibility and applicability across diverse deployment scenarios.

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